Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Remove Even Numbers from a Set

Write a program to create a set of integers and remove all the even numbers from the set. Display the updated set.

Q2. Check Element in Tuple

Create a tuple containing 6 animal names. Accept an animal name from the user and check whether it is present in the tuple.

Q3. Tuple Slicing

Write a Python program to accept 8 numbers from the user and store them in a tuple. Using tuple slicing, display:

1. The first four elements
2. The last three elements
3. The tuple in reverse order

Q4. Tuple Concatenation

Write a Python program to accept 3 electronic device names from the user for the first tuple and 3 different electronic device names for the second tuple. Combine the two tuples using the + operator and display the first tuple, second tuple, and the resulting combined tuple.

Q5. Common Items in Two Shopping Lists

Write a Python program to accept 5 item names from the user for the first shopping list and 5 item names for the second shopping list. Convert both lists into sets and display:

1. Items that are common to both shopping lists.
2. Items present only in the first shopping list.
3. Items present only in the second shopping list.

Q6. Tuple Element Frequency

Write a program to create a tuple of integers and display the frequency of each element present in the tuple.

Q7. Programming Language Search Using Tuple

Write a Python program to accept the names of 6 programming languages from the user and store them in a tuple. Then accept another programming language from the user and search for it in the tuple using the `index()` method. If the language is present, display its position; otherwise, display an appropriate message.

Q8. Unique Characters in a Sentence

Accept a sentence from the user and create a set containing all the **unique characters** in the sentence, excluding spaces. Display the set and the total number of unique characters.

Solutions

1. Remove Even Numbers from a Set

```
# Create a set
numbers = {10, 15, 20, 25, 30, 35, 40}

# Create an empty set for odd numbers
updated_set = set()

# Add only odd numbers to the new set
for num in numbers:
    if num % 2 != 0:
        updated_set.add(num)

print("Original Set:", numbers)
print("Updated Set:", updated_set)
```

Output

Original Set: {35, 20, 40, 25, 10, 30, 15}
Updated Set: {35, 25, 15}

2. Check Element in Tuple

```
animals = ()

for i in range(6):
    animal = input("Enter animal name " + str(i + 1) + ": ")
    animals += (animal,)

search_animal = input("Enter the animal to search: ")

if search_animal in animals:
    print("Animal is present in the tuple")
else:
    print("Animal is not present in the tuple")
```

Output 1

```
Enter animal name 1: Lion
Enter animal name 2: Tiger
Enter animal name 3: Elephant
Enter animal name 4: Horse
Enter animal name 5: Dog
Enter animal name 6: Cat
Enter the animal to search: Tiger

Animal is present in the tuple
```

Output 2

```
Enter animal name 1: Lion
Enter animal name 2: Tiger
Enter animal name 3: Elephant
Enter animal name 4: Horse
Enter animal name 5: Dog
Enter animal name 6: Cat
Enter the animal to search: Rabbit

Animal is not present in the tuple
```

3. Tuple Slicing

```
numbers = ()

for i in range(8):
```

```
    number = int(input("Enter number " + str(i + 1) + ": "))
    numbers += (number,)

print("Tuple:", numbers)
print("First four elements:", numbers[:4])
print("Last three elements:", numbers[-3:])
print("Reverse order:", numbers[::-1])
```

Output

```
Enter number 1: 10
Enter number 2: 20
Enter number 3: 30
Enter number 4: 40
Enter number 5: 50
Enter number 6: 60
Enter number 7: 70
Enter number 8: 80

Tuple: (10, 20, 30, 40, 50, 60, 70, 80)
First four elements: (10, 20, 30, 40)
Last three elements: (60, 70, 80)
Reverse order: (80, 70, 60, 50, 40, 30, 20, 10)
```

4. Tuple Concatenation

```
tuple1 = ()
tuple2 = ()

print("Enter 3 electronic devices for the first tuple:")

for i in range(3):
    device = input("Enter device " + str(i + 1) + ": ")
    tuple1 += (device,)

print("\nEnter 3 electronic devices for the second tuple:")

for i in range(3):
    device = input("Enter device " + str(i + 1) + ": ")
    tuple2 += (device,)
```

```
combined_tuple = tuple1 + tuple2

print("\nFirst tuple:", tuple1)
print("Second tuple:", tuple2)
print("Combined tuple:", combined_tuple)
```

Output

```
Enter 3 electronic devices for the first tuple:
Enter device 1: Laptop
Enter device 2: Keyboard
Enter device 3: Mouse

Enter 3 electronic devices for the second tuple:
Enter device 1: Mobile
Enter device 2: Printer
Enter device 3: Scanner

First tuple: ('Laptop', 'Keyboard', 'Mouse')
Second tuple: ('Mobile', 'Printer', 'Scanner')
Combined tuple: ('Laptop', 'Keyboard', 'Mouse', 'Mobile', 'Printer', 'Scanner')
```

5. Common Items in Two Shopping Lists

```
list1 = []
list2 = []

print("Enter 5 items for the first shopping list:")

for i in range(5):
    item = input("Enter item " + str(i + 1) + ": ")
    list1.append(item)

print("\nEnter 5 items for the second shopping list:")

for i in range(5):
    item = input("Enter item " + str(i + 1) + ": ")
    list2.append(item)

set1 = set(list1)
set2 = set(list2)
```

```
print("\nCommon items:", set1 & set2)
print("Items only in first list:", set1 - set2)
print("Items only in second list:", set2 - set1)
```

Output

```
Enter 5 items for the first shopping list:
Enter item 1: Rice
Enter item 2: Milk
Enter item 3: Bread
Enter item 4: Apples
Enter item 5: Soap

Enter 5 items for the second shopping list:
Enter item 1: Milk
Enter item 2: Bread
Enter item 3: Eggs
Enter item 4: Juice
Enter item 5: Shampoo

Common items: {'Milk', 'Bread'}
Items only in first list: {'Rice', 'Apples', 'Soap'}
Items only in second list: {'Eggs', 'Juice', 'Shampoo'}
```

6. Tuple Element Frequency

```
t = (10, 20, 30, 20, 40, 10, 50, 30, 20)

visited = ()

print("Element\tFrequency")

for num in t:
    if num not in visited:
        print(num, "\t", t.count(num))
        visited += (num,)
```

Output

```
Element  Frequency
10 2
```

20 3
30 2
40 1
50 1

7. Programming Language Search Using Tuple

```
  languages = ()

for i in range(6):
    language = input("Enter programming language " + str(i + 1) + ": ")
    languages += (language,)

search_language = input("Enter the language to search: ")

if search_language in languages:
    position = languages.index(search_language)
    print("Language found at position:", position + 1)
else:
    print("Language not found")
```

Output1 (Language found case)

```
Enter programming language 1: Python
Enter programming language 2: C
Enter programming language 3: Java
Enter programming language 4: C++
Enter programming language 5: JavaScript
Enter programming language 6: Ruby
Enter the language to search: Java

Language found at position: 3
```

Output 2(Language not found case)

```
Enter programming language 1: Python
Enter programming language 2: C
Enter programming language 3: Java
Enter programming language 4: C++
Enter programming language 5: JavaScript
Enter programming language 6: Ruby
```

Enter the language to search: PHP

Language not found

8. Unique Characters in a Sentence

```
sentence = input("Enter a sentence: ")

characters = set()

for ch in sentence:
    if ch != " ":
        characters.add(ch)

print("Unique characters:", characters)
print("Number of unique characters:", len(characters))
```

Output 1

Enter a sentence: Python Programming

Unique characters: {'P', 'y', 't', 'h', 'o', 'n', 'p', 'r', 'g', 'a', 'm', 'i'}
Number of unique characters: 12